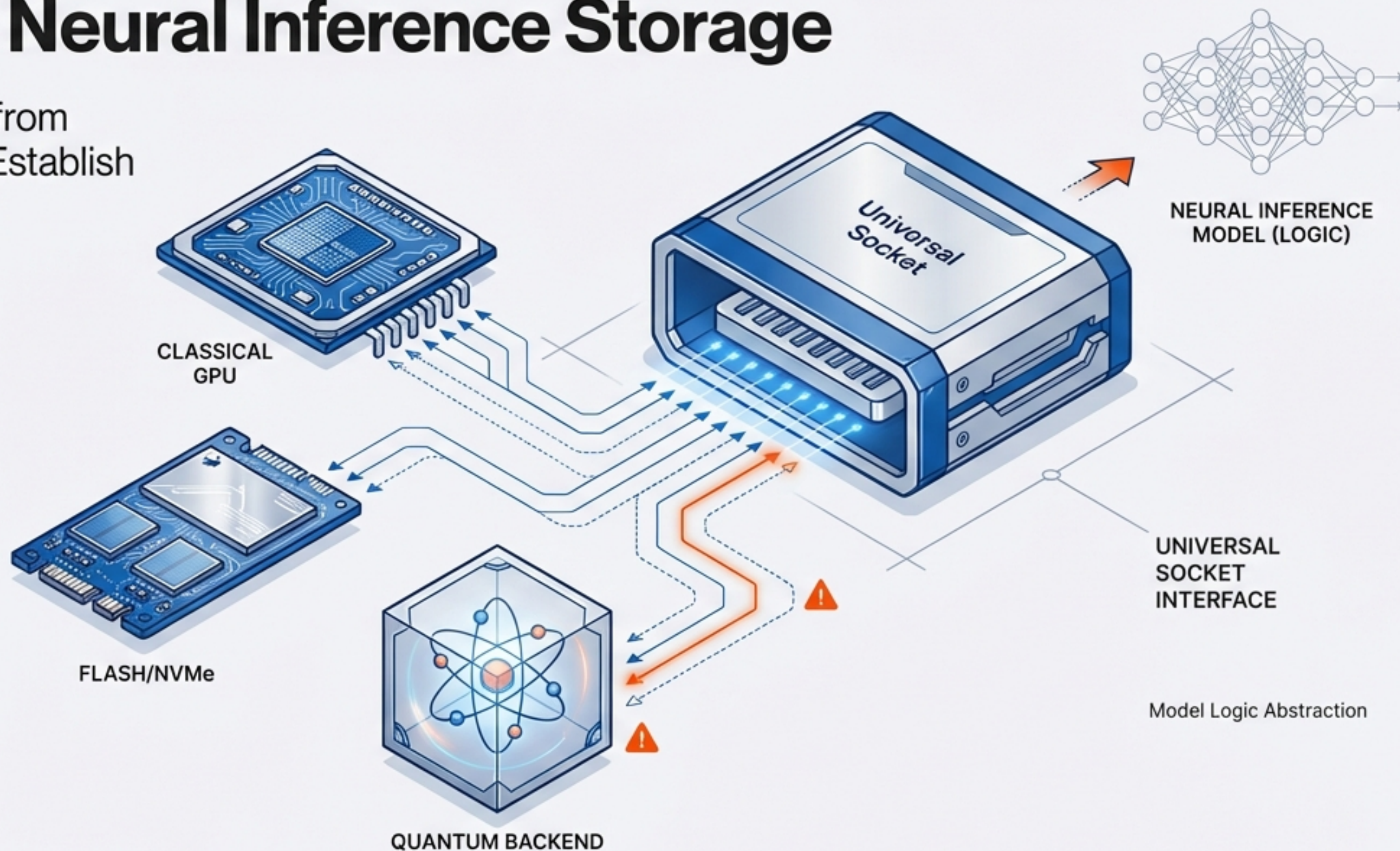


Memory Socket: The USB for Neural Inference Storage

Decoupling Model Logic from
Memory Architecture to Establish
a New Research Vertical.



The Context Window Paradox

Infinite Inputs, Finite Throughput

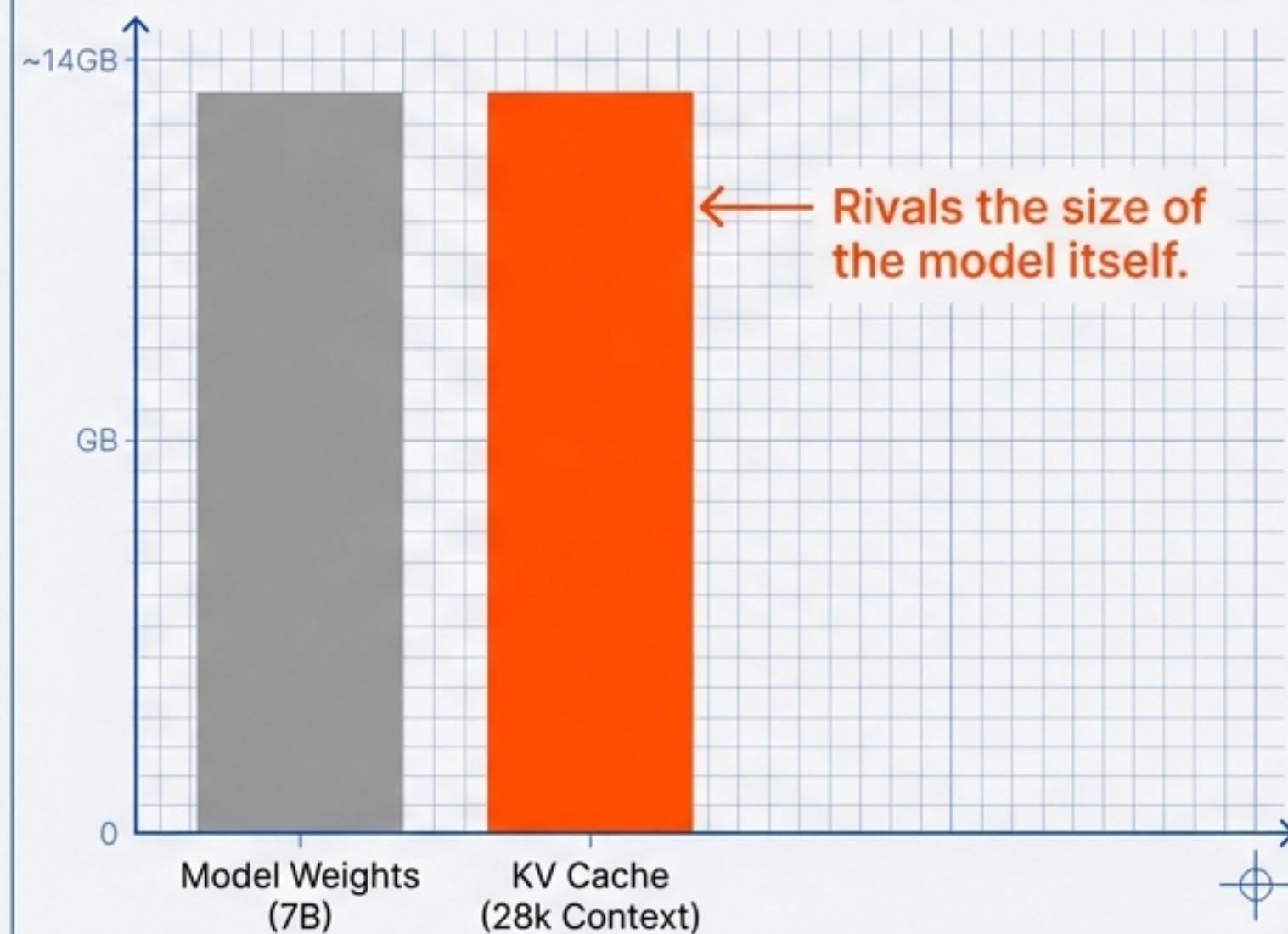
While context windows explode (GPT-5: 256k, Claude: 200k), memory management remains the critical bottleneck.

The Math of the Bottleneck:

- 7B Parameter Model: ≈ 0.5 MB KV-cache per token.
- Single 28k Prompt: ≈ 14 GB cache size.
- 175B Scale Model: ≈ 4 MB per token.

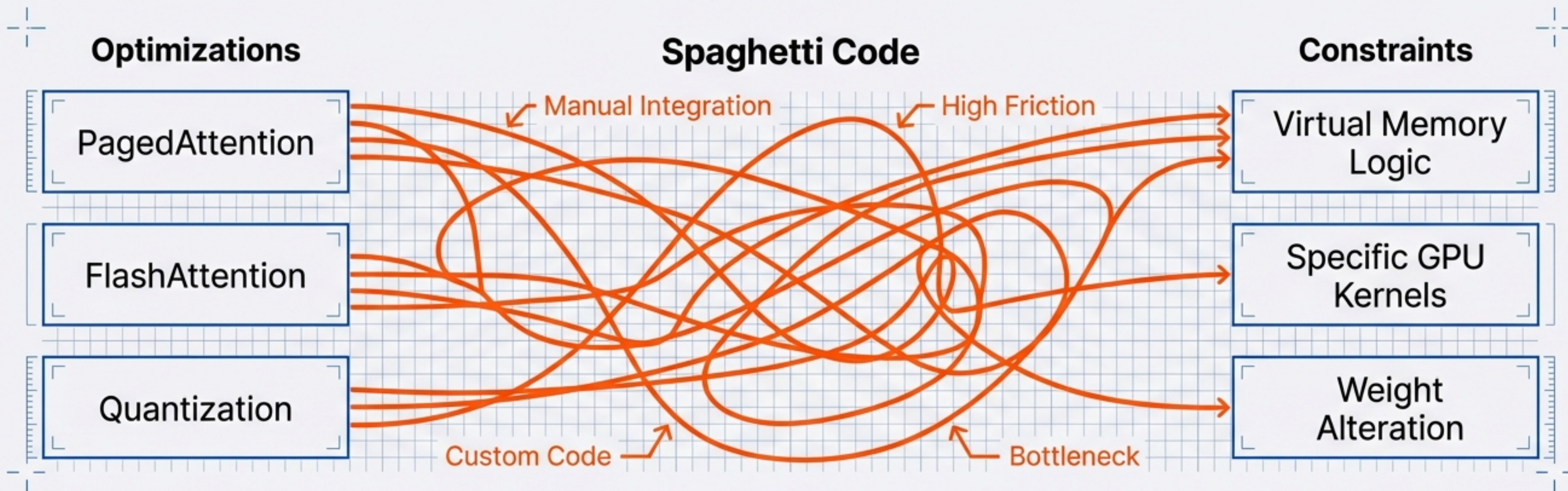
Result: Beyond ~ 8 k tokens, inference is memory-bandwidth bound, capping GPU utilization at 35–45%.

The Hidden Cost of Context



The Pre-Standardization Era

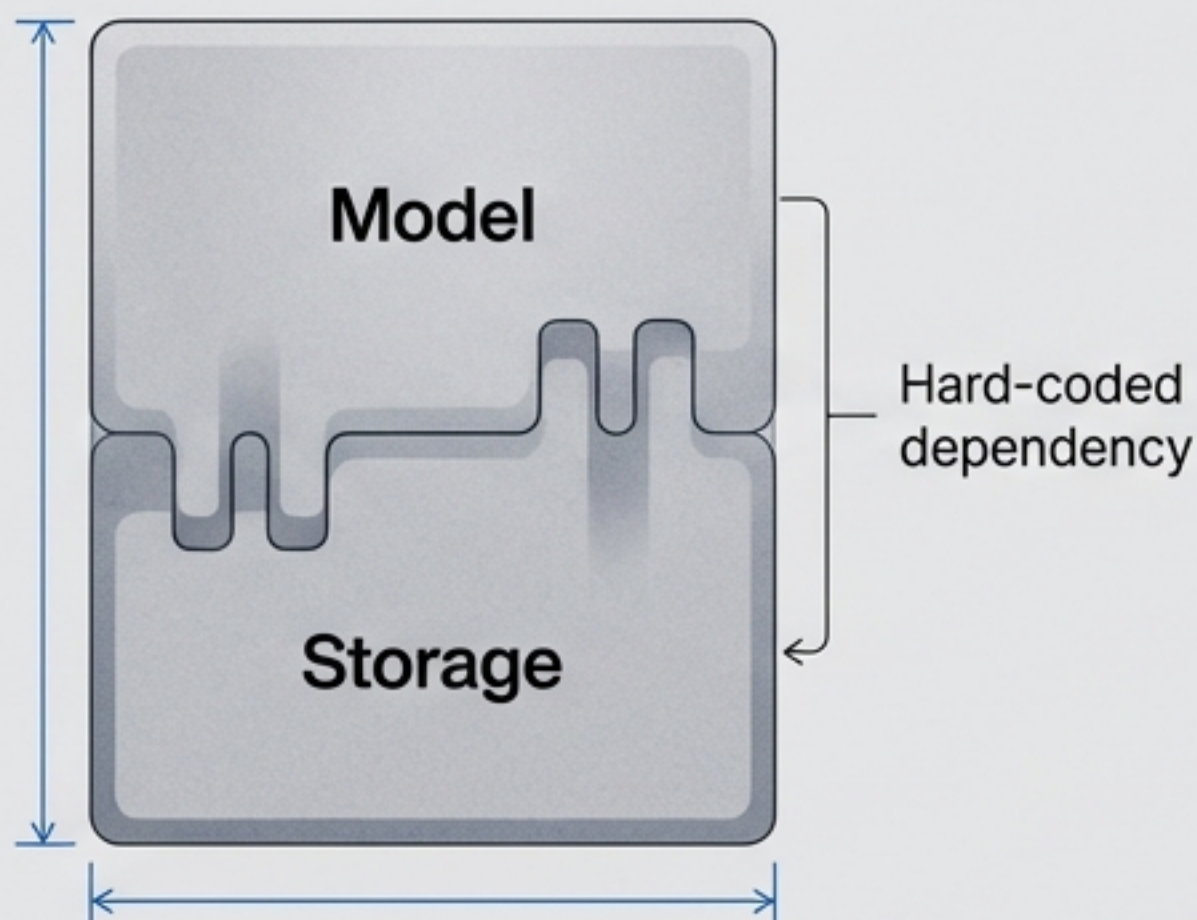
A Fragmented Ecosystem (The Storage Abstraction Void)



The Consequence: Every new storage idea requires rewriting the model integration logic. There is no standard interface.

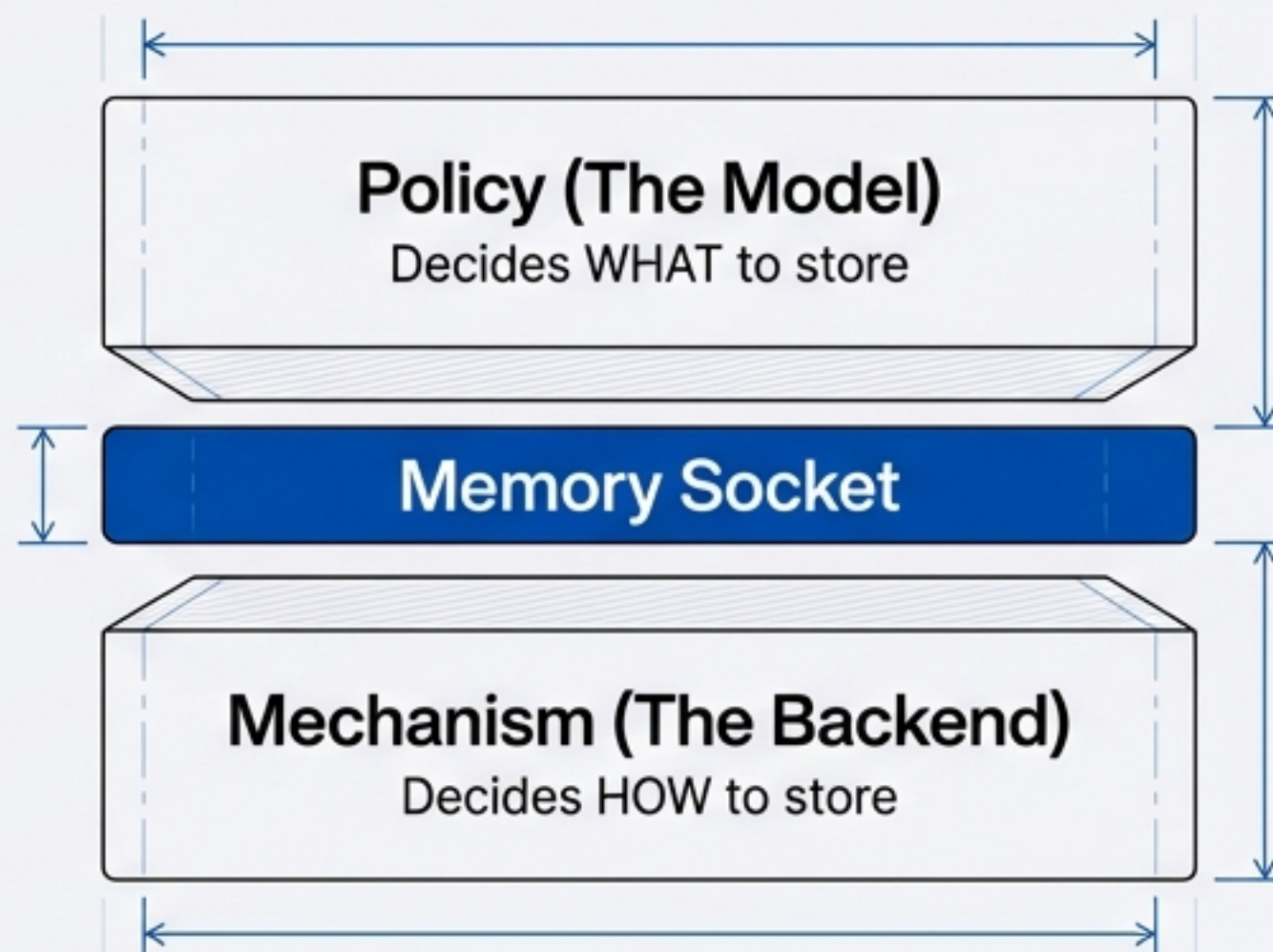
The Abstraction: Decoupling Policy from Mechanism

Monolithic Architecture



- Changing storage requires changing the model.

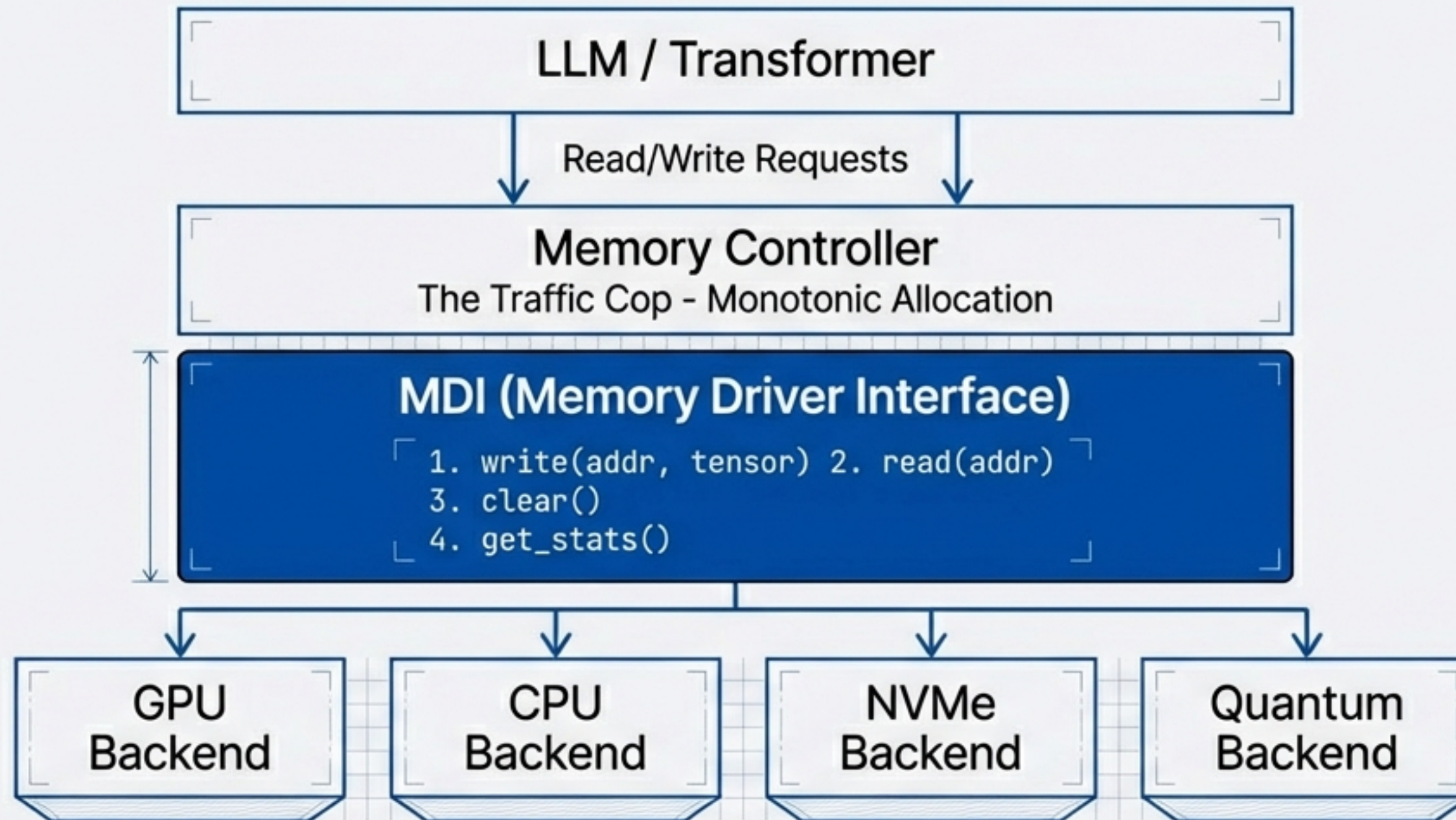
Decoupled Architecture



Just as USB decoupled peripherals from the motherboard, Memory Socket decouples inference memory from the LLM.

System Architecture

The Memory Driver Interface (MDI)



Any storage medium that satisfies these four commands can drive an LLM.

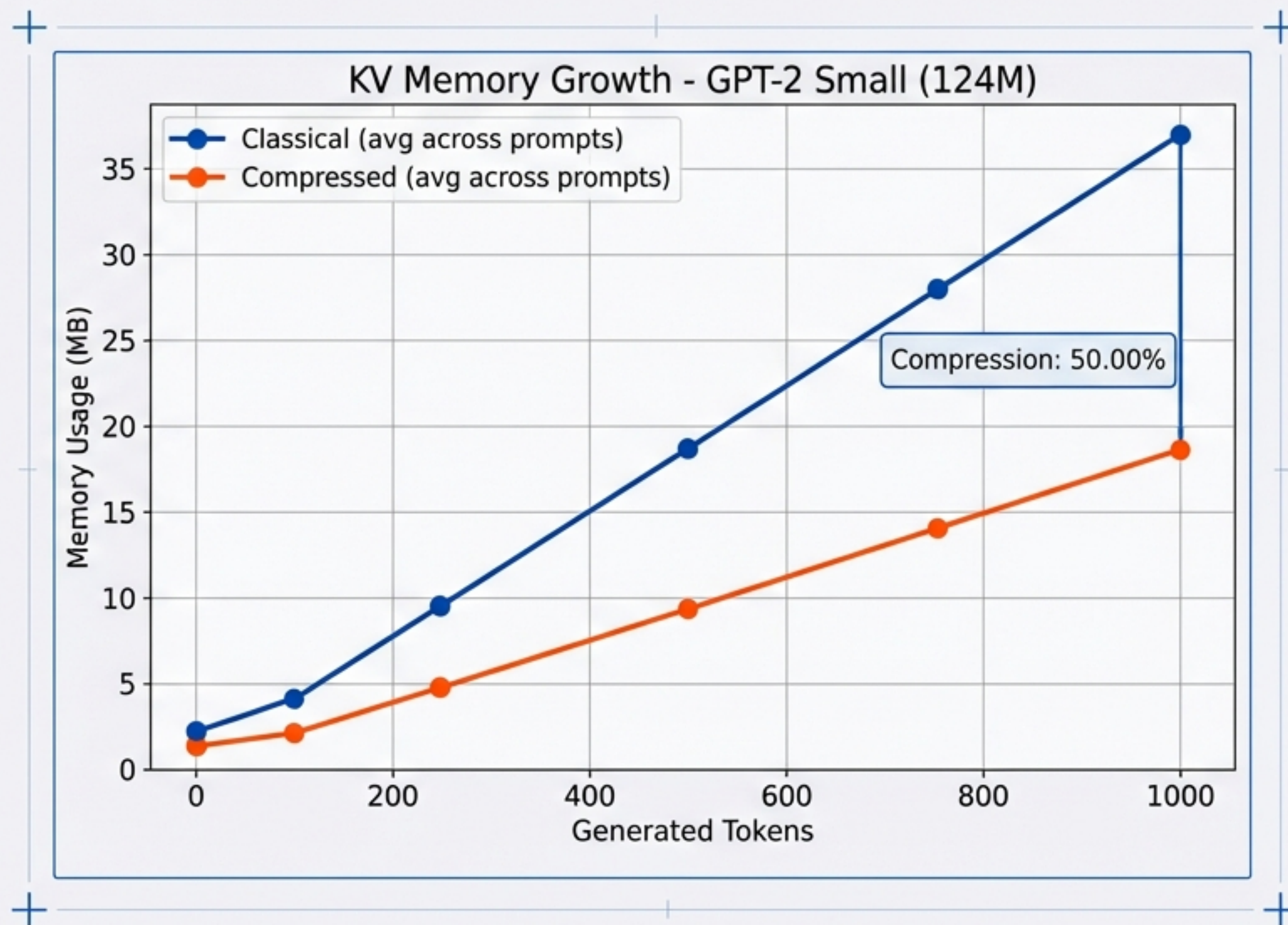
Validating the Ecosystem

2x Compression with Zero Latency

Experiment: Comparing Backend-Class (Reference) vs. Backend-Comp (Int8/16).

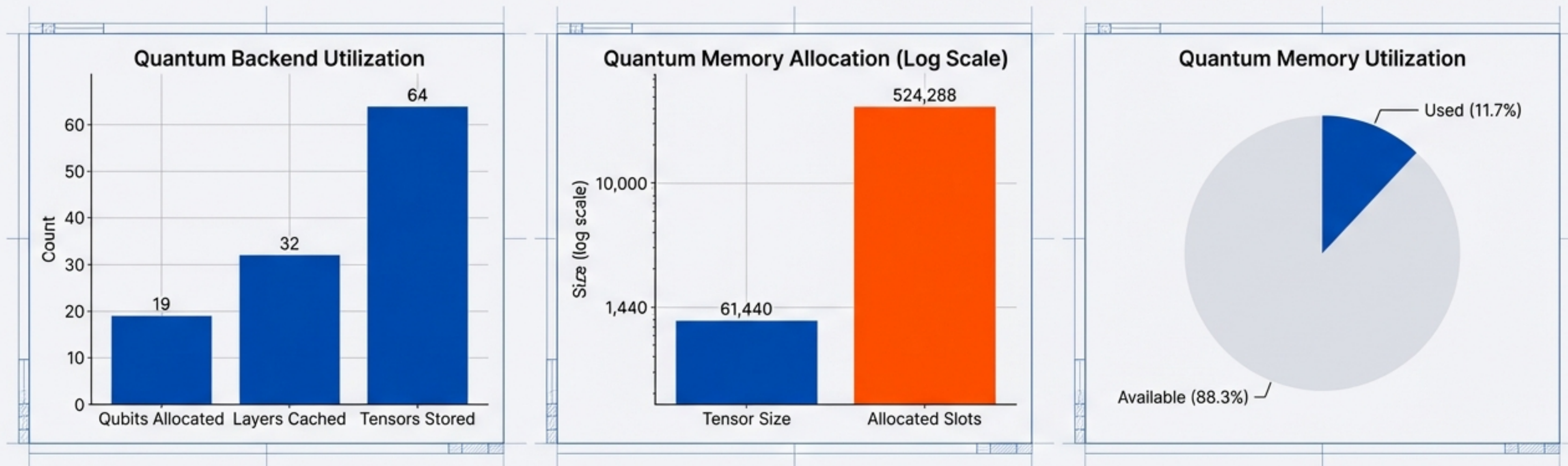
Results:

- Memory: Exact 2.00x reduction.
- Throughput: Zero penalty (<1% variance).
- Fidelity: Bit-exact generation.



The Ultimate Stress Test: Quantum Storage

Proving Transformers are Storage-Agnostic



Neue Haas Grotesk Display Pro Inter

Status: VALIDATED. Text Generated: Identical to Classical Baseline.

From Optimization to Diagnostics

Establishing the First Neural Memory Benchmark

Turn 1: **Implantation**



User provides a
secret code.



Turn 2: **Distraction**



Discussion of **distraction**
topic (Physics/History) to
force **context switching**.



Turn 3: **Recall**

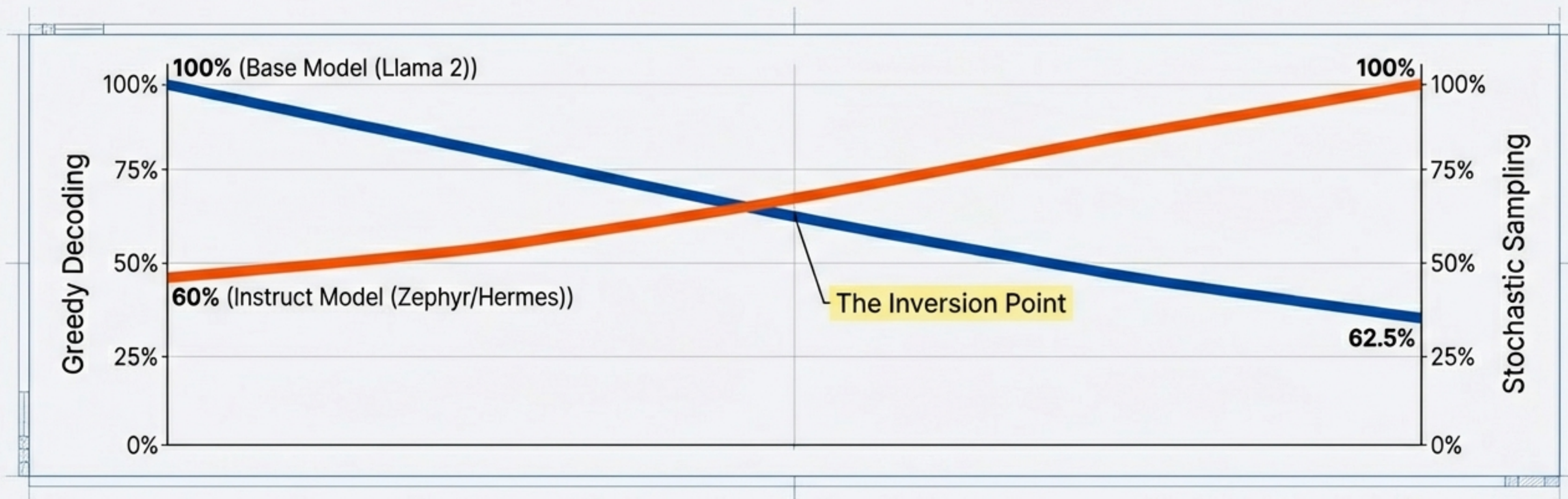


User **asks for** the code.
Model must **retrieve**
from backend.

Goal: Measure 'Statefulness' and 'Recall' systematically across architectures and storage backends.

Discovery 1: The Instruction Tuning Inversion

How Tuning Fundamentally Alters Memory Behavior



Base models prioritize exact recall under Greedy decoding. Instruction-tuned models fail at Greedy but achieve perfect recall under Stochastic sampling.

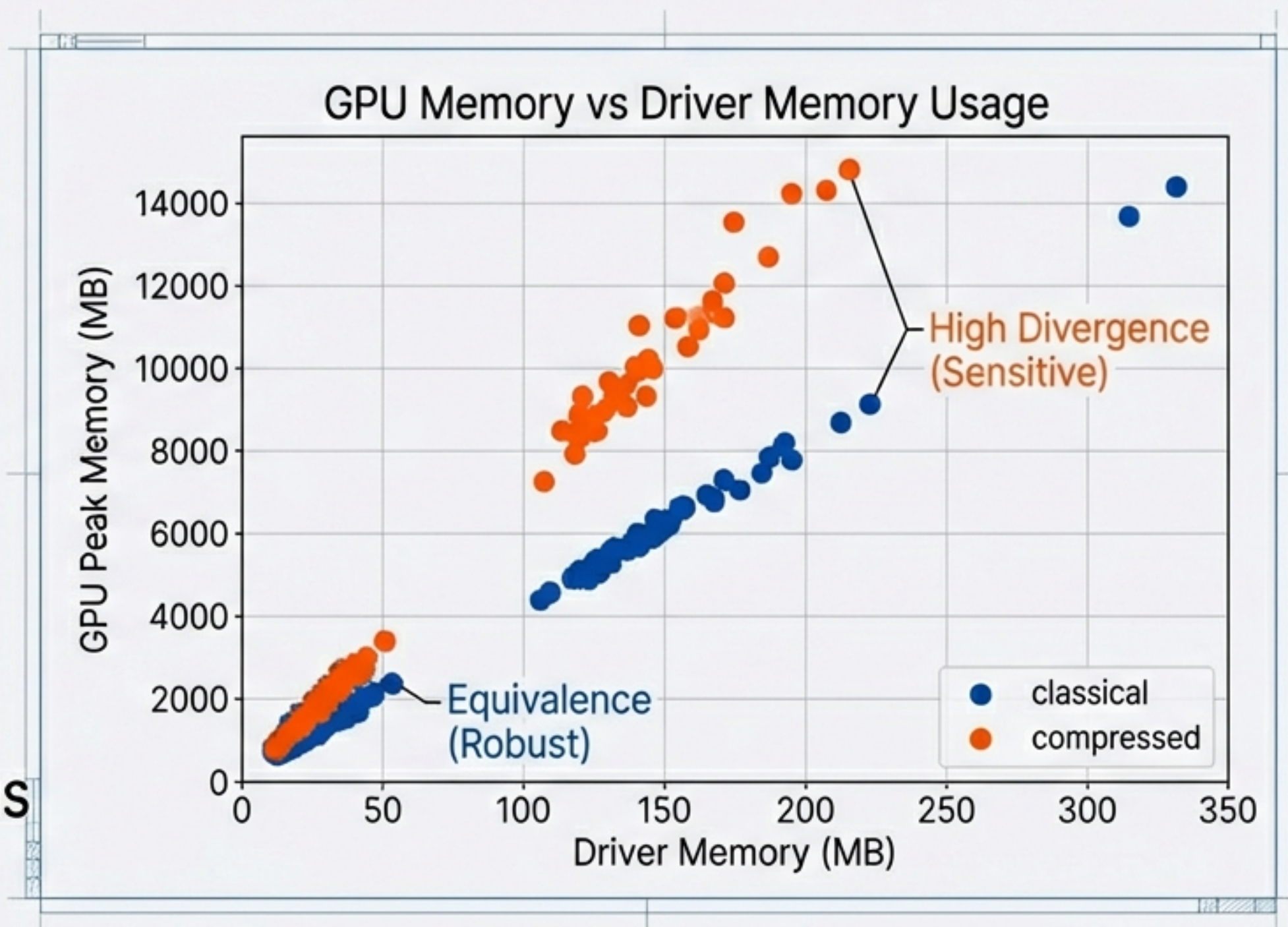
Discovery 2: Backend Divergence as Diagnostics

Using Incompatibility as a Signal

Observation: 77.8% of model configurations showed behavior **divergence** between Classical and Compressed backends.

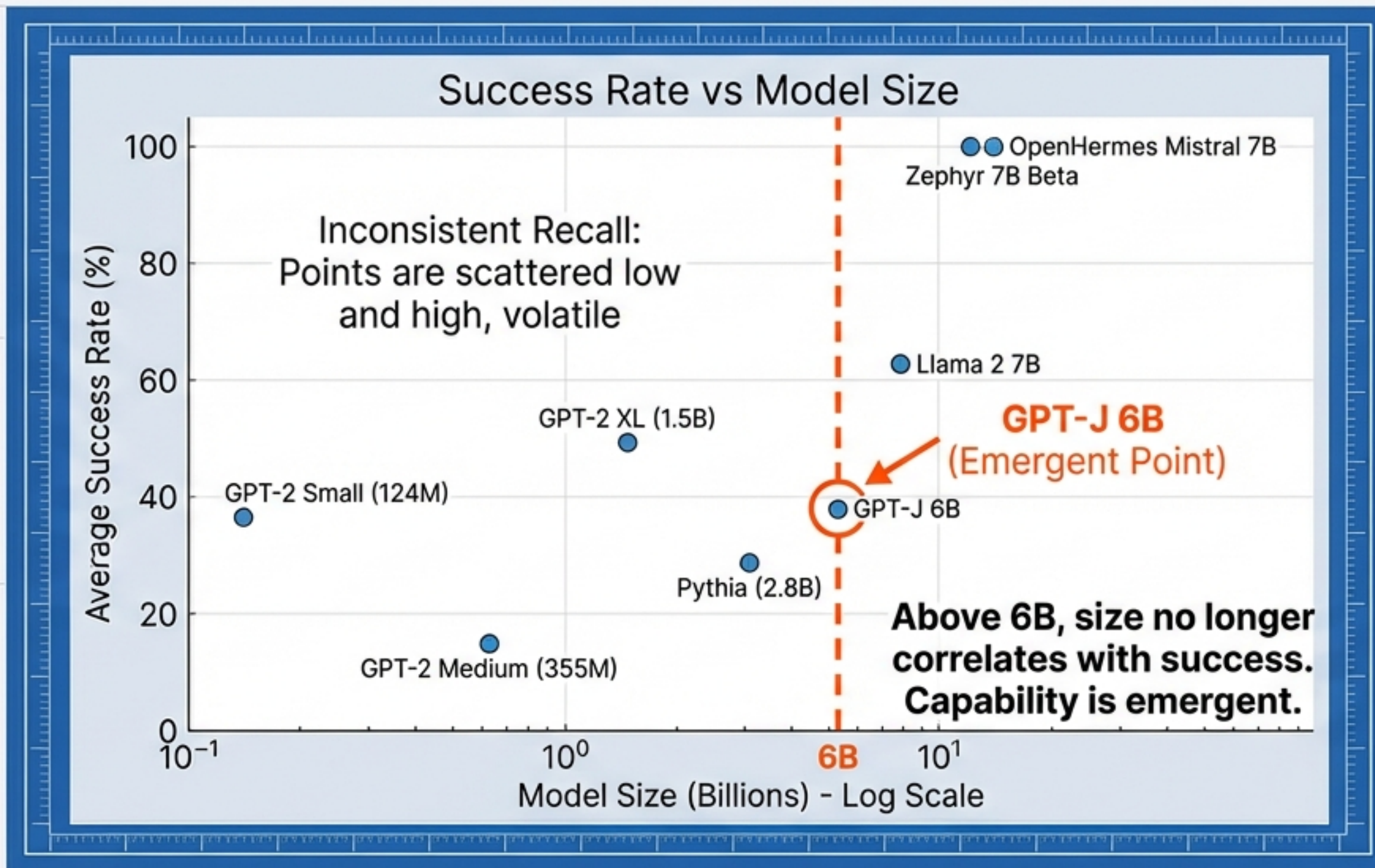
Insight: High **divergence** indicates fragility to quantization. Low divergence indicates **robust** memory representations.

Takeaway: The MDI acts as a litmus test for deployment safety.



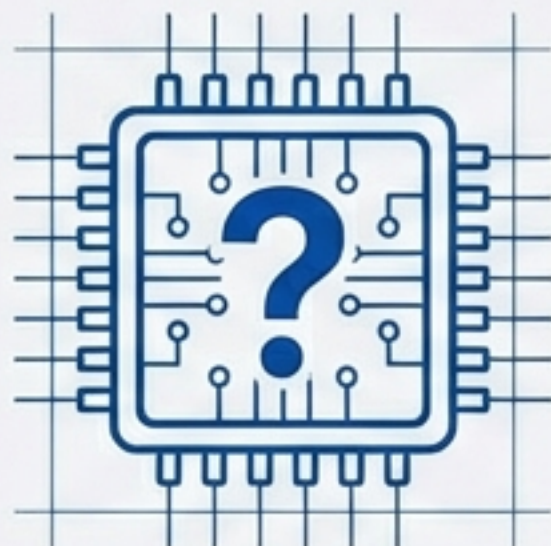
Discovery 3: The 6 Billion Parameter Threshold

Memory Capability Emerges at Scale



A New Research Vertical

What Memory Socket Enables



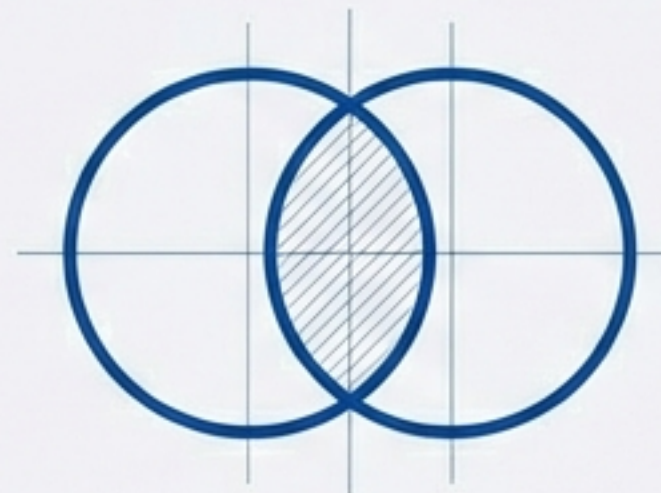
Hardware Co-design

Test novel chips (**PIM**, **QRAM**) immediately using a simple **Python driver**. No kernel rewriting required.



Tiered Storage Logic

Implement complex hierarchies (GPU > CPU > NVMe) entirely within the backend **`write`** logic, logic, invisible to the model.



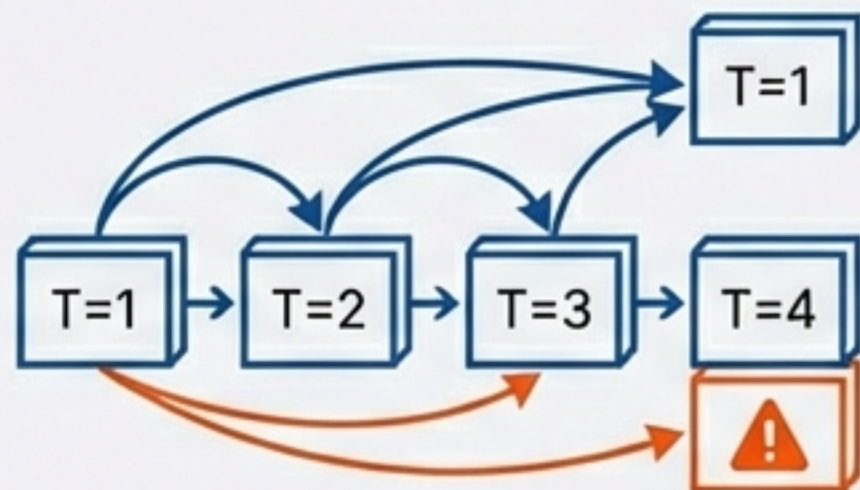
Cross-Pollination

Storage engineers optimize backends without NLP expertise.
NLP researchers use advanced storage without kernel expertise.

Roadmap & Engineering Frontiers

Current Constraints

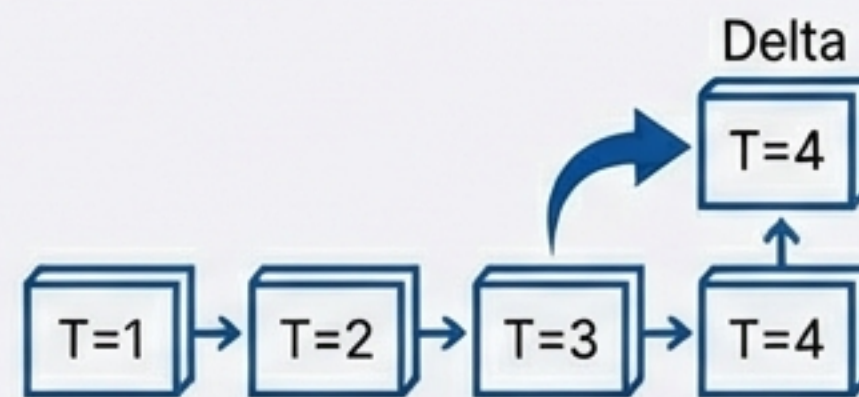
Reference Controller uses $O(T^2)$ writes (full concatenation).



Bottleneck: $O(T^2)$ Writes

Solution:

Backend-specific Delta updates (append-only) via the same 'MDI'.



Delta Update: Append-only, $O(1)$ for new data.

Future Directions

True Infinite Context



Moving from paging to streaming via dedicated hardware backends. Data flows seamlessly without blocking.

Real Quantum RAM



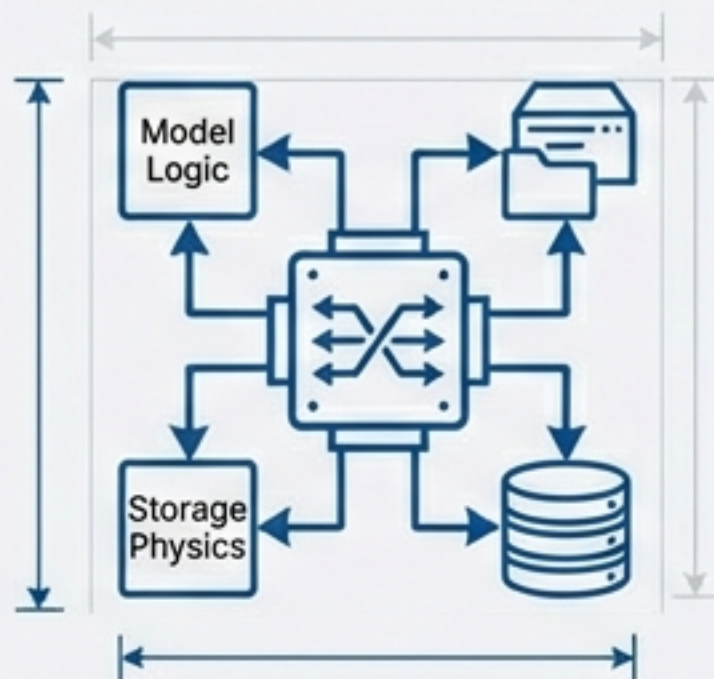
Transitioning from simulation to $O(\sqrt{N})$ lookup on quantum hardware. Exponential speedup for data retrieval.

Protocol Expansion



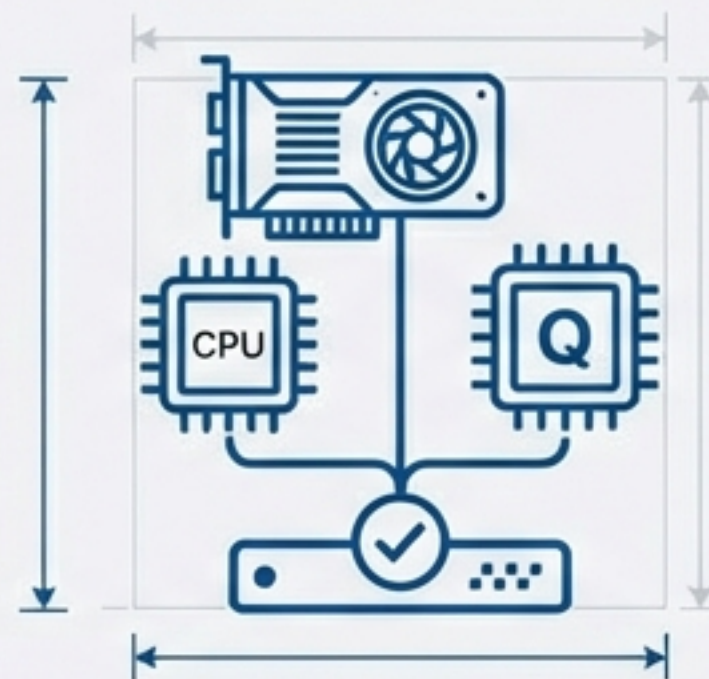
Extending the benchmark to code generation and legal analysis tasks. Broadening the scope of evaluation.

Summary: A New Paradigm for Inference



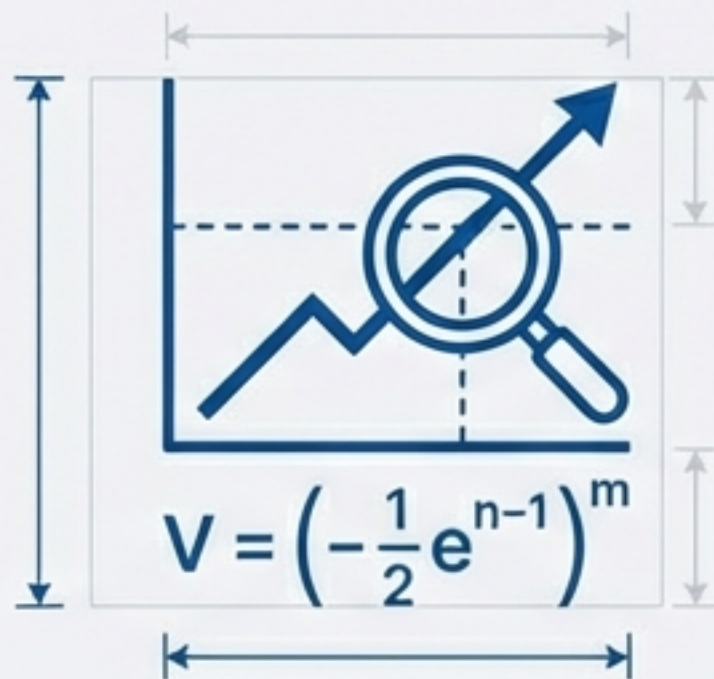
Universal Interface

Decoupled Model Logic from Storage Physics via MDI.



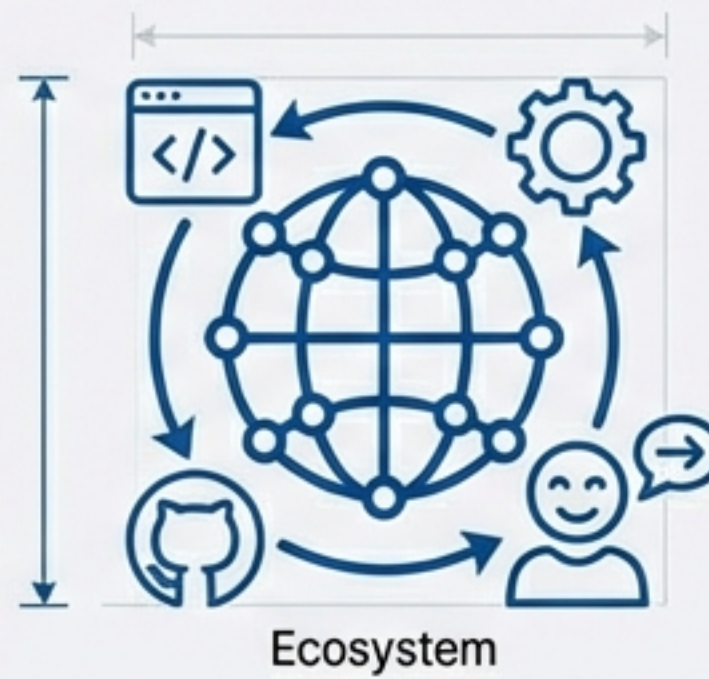
Validated Agnosticism

Proved transformers work on GPU, CPU, and Quantum substrates identically.



Scientific Insight

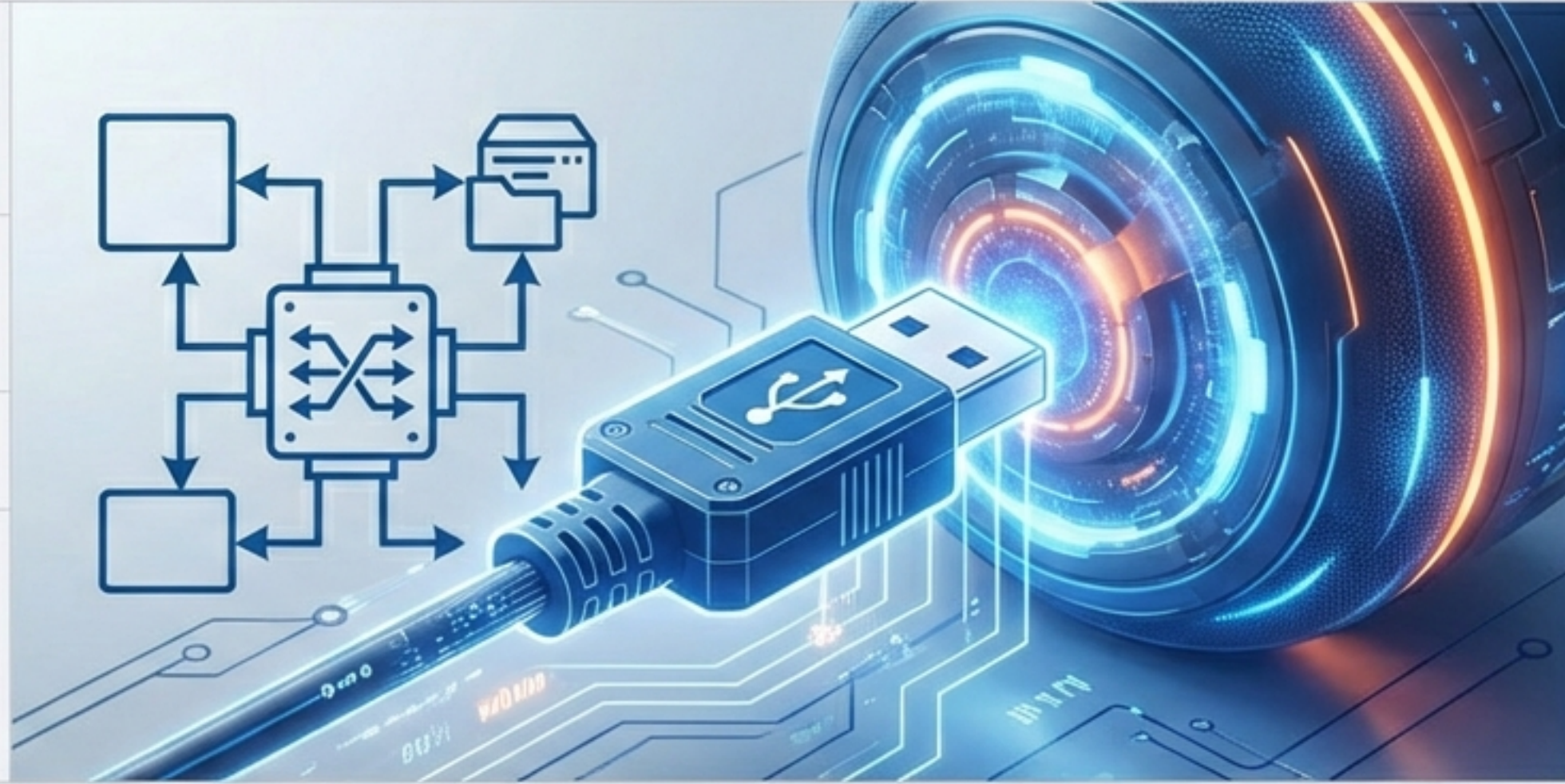
Revealed Instruction Tuning Inversion & 6B Parameter Threshold.



Open Source

Infrastructure available now to accelerate the ecosystem.

Future-Proofing Neural Inference



“The Memory Socket transforms every future storage innovation into a compatible backend.”

Standardization precedes explosion. The era of bespoke memory integration is over.